

TURAN, Russian Federation

WMO: 360920

Lat: 52.1451N

Lon: 93.9159E

Elev: 2797

StdP: 13.27

Time Zone: 7.00 (E07)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-41.9	-37.7	-48.7	0.3	-41.6	-44.2	0.5	-37.7	9.4	3.3	6.9	-7.7	0.3	250	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.4	87.2	59.2	83.1	58.6	79.5	57.9	62.9	77.1	61.4	75.8	60.1	74.2	5.2	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.3	80.6	66.7	56.7	75.9	65.2	55.2	71.9	63.9	29.9	77.1	28.8	75.3	27.8	74.8	70.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.2	10.9	9.2	DB	-44.7	92.9	3.8	3.9	-47.4	95.7	-49.7	98.0	-51.8	100.2	-54.6	103.0	(4)
(5)			WB	-44.8	65.1	3.7	1.4	-47.5	66.1	-49.7	66.9	-51.8	67.6	-54.6	68.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	28.3	-19.4	-9.9	16.8	38.9	48.4	62.6	64.9	60.7	47.9	31.8	6.0	-12.0	(6)	
	DBStd	30.96	9.62	11.08	12.74	8.27	8.70	7.55	5.24	5.91	8.16	8.83	13.09	10.10	(7)	
	HDD50	9285	2151	1678	1029	342	138	6	0	2	135	564	1319	1921	(8)	
	HDD65	13559	2616	2098	1494	784	518	130	64	159	513	1028	1769	2386	(9)	
	CDD50	1348	0	0	0	8	87	383	463	334	72	1	0	0	(10)	
	CDD65	147	0	0	0	0	2	58	62	25	0	0	0	0	(11)	
	CDH74	2291	0	0	0	1	112	940	782	406	50	0	0	0	(12)	
	CDH80	706	0	0	0	0	20	321	248	109	8	0	0	0	(13)	
	WSAvg	3.5	1.7	2.4	3.5	5.2	5.4	4.6	4.2	3.9	3.9	3.3	2.4	1.7	(14)	
	PrecAvg	28.5	1.0	1.0	1.5	1.9	2.4	3.1	4.1	4.0	2.7	2.7	2.4	2.0	(15)	
(16) Precipitation	PrecMax	37.6	2.4	3.2	4.3	4.0	5.6	5.1	9.6	8.0	4.3	12.0	5.8	6.1	(16)	
	PrecMin	19.3	0.1	0.0	0.2	0.5	0.0	0.6	0.2	2.2	1.4	0.6	0.7	0.1	(17)	
	PrecStd	4.4	0.6	0.7	1.0	1.1	1.4	1.2	1.5	1.3	0.8	2.2	1.3	1.5	(18)	
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.9	20.7	50.1	71.5	82.3	92.4	92.7	88.8	79.5	63.1	37.1	20.6	(19)	
		MCWB	7.4	18.6	37.9	47.8	55.3	59.4	60.8	59.5	55.4	46.5	31.8	19.2	(20)	
	2%	DB	2.6	14.7	43.4	64.8	77.1	87.4	86.9	82.9	73.6	55.9	32.5	13.6	(21)	
		MCWB	1.5	13.2	34.3	45.6	52.6	58.7	60.4	60.0	54.8	43.3	29.5	12.6	(22)	
	5%	DB	-1.1	10.9	38.4	59.8	71.5	83.6	82.1	78.4	68.6	50.0	28.8	7.4	(23)	
		MCWB	-1.9	9.2	31.7	43.2	50.7	57.8	59.6	59.2	52.8	40.8	26.4	6.4	(24)	
	10%	DB	-5.0	7.2	34.0	54.9	66.4	79.6	78.1	74.4	63.2	44.9	24.5	2.0	(25)	
		MCWB	-5.7	5.9	29.6	41.1	47.8	57.1	59.2	57.9	50.1	37.6	22.3	1.3	(26)	
	0.4%	WB	7.7	18.5	38.4	49.4	55.8	63.0	64.7	65.1	57.5	47.8	32.5	19.2	(27)	
		MCDB	9.0	20.7	48.5	66.7	77.8	79.6	78.4	77.2	72.9	59.0	36.0	20.7	(28)	
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	1.6	13.0	34.8	46.6	53.8	61.3	63.1	62.7	55.4	44.5	29.5	12.8	(29)	
		MCDB	2.5	14.5	42.3	60.8	73.9	78.2	77.3	75.4	71.3	54.7	31.9	13.8	(30)	
	5%	WB	-1.9	9.4	32.2	43.9	51.7	60.1	61.8	60.6	53.5	41.4	26.3	6.6	(31)	
		MCDB	-1.2	10.8	37.3	57.5	69.7	76.8	75.8	74.0	67.5	48.8	28.6	7.6	(32)	
	10%	WB	-5.7	6.1	29.7	41.7	49.0	58.8	60.6	59.0	51.0	38.6	22.3	1.4	(33)	
		MCDB	-5.0	7.4	33.9	54.6	63.7	75.2	74.2	71.6	61.7	44.2	24.3	2.1	(34)	
(36) Mean Daily Temperature Range	5% DB	MDBR	16.6	20.4	20.4	22.9	24.9	25.7	22.4	23.0	24.4	20.1	16.9	14.7	(35)	
		MCDBR	18.1	20.8	21.6	31.1	33.3	32.3	30.1	30.6	32.5	28.6	16.5	14.5	(36)	
		MCWBR	17.1	19.4	14.1	15.7	14.7	10.8	9.9	11.3	14.9	16.7	13.3	13.6	(37)	
	5% WB	MCDBR	17.9	20.7	19.6	27.0	30.0	26.0	22.9	24.4	30.7	27.4	16.2	14.6	(38)	
		MCWBR	16.9	19.3	13.3	14.3	13.8	9.9	9.1	10.3	14.5	16.3	13.2	13.7	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.263	0.294	0.339	0.379	0.390	0.385	0.383	0.369	0.334	0.320	0.294	0.267	(40)	
	taud		2.105	2.068	2.017	2.169	2.230	2.350	2.388	2.397	2.434	2.348	2.170	2.060	(41)	
	Ebn at Noon		228	255	265	270	273	276	274	272	270	247	213	200	(42)	
	Edh at Noon		23	33	43	42	42	37	35	33	28	25	22	21	(43)	
(44) All-Sky Solar Radiation	RadAvg		403	752	1243	1548	1742	1924	1753	1509	1136	692	375	280	(44)	
	RadStd		22	37	52	72	121	156	109	107	112	56	36	19	(45)	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page